

Part A: Classifying Numbers

Classify each number using all that apply: rational, irrational, whole, integer.

1. 0

5. 4

2. $\sqrt{81}$

6. $\sqrt{64}$

3. $\sqrt{7}$

7. $\sqrt{36}$

4. 2

8. $\frac{2}{11}$

Part B: Negative Number Operations

Evaluate each expression completely.

1. $6 - 5$

5. $84 \div (-12) - 6$

2. $6 - (-11)$

6. $6 \div 3 - (-4)$

3. $45 \div (-9)$

7. $3^3 - (-17) \cdot 3$

4. $-36 \div 9$

8. $(-3 + (-11))^2 \div 1$

Part C: Decimals

Evaluate. Write each answer as a decimal.

1. $14.5 + 9.7$

3. $14.55 \div 0.5$

2. 9.6×1.8

Part D: Decimals to Fractions

Write each decimal as a fraction in lowest terms.

1. 11.3

Part E: Fractions*Evaluate. Write every answer in lowest terms.*

1. $\frac{1}{2} + \frac{3}{5}$

3. $5\frac{1}{5} + 6$

2. $\frac{1}{2} \div \frac{1}{4}$

Part F: Writing Fractions in Lowest Terms*Write each fraction in lowest terms.*

1. $\frac{20}{32}$

Part G: Exponent Rules*Simplify. Write answers with positive exponents.*

1. $3x^4 \cdot 4x^3$

5. $(4x^8)^2$

2. $x^4 \cdot 4x^6$

6. $2 \cdot 2^{-2}$

3. $\frac{x^4}{x^2}$

7. $6^2 \cdot 6^{-4}$

4. $\frac{x^7}{x^5}$

8. $\left(\frac{3x^8}{x^4}\right)^2$

Answer Key

Part A: Classifying Numbers

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|-----------------------------|-----------------------------|-----------------------------|
| 1. Rational, Integer, Whole | 4. Rational, Integer, Whole | 7. Rational, Integer, Whole |
| 2. Rational, Integer, Whole | 5. Rational, Integer, Whole | 8. Rational |
| 3. Irrational | 6. Rational, Integer, Whole | |

Part B: Negative Number Operations

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| 1. 1 | 4. -4 | 7. 78 |
| 2. 17 | 5. -13 | 8. 196 |
| 3. -5 | 6. 6 | |

Part C: Decimals

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| 1. 24.2 | 2. 17.28 | 3. 29.1 |
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Part D: Decimals to Fractions

1. $\frac{113}{10}$

Part E: Fractions

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|--------------------|------|--------------------|
| 1. $1\frac{1}{10}$ | 2. 2 | 3. $11\frac{1}{5}$ |
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Part F: Writing Fractions in Lowest Terms

1. $\frac{5}{8}$

Part G: Exponent Rules

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|--------------|------------------|-------------------|
| 1. $12x^7$ | 4. x^2 | 7. $\frac{1}{36}$ |
| 2. $4x^{10}$ | 5. $16x^{16}$ | 8. $9x^8$ |
| 3. x^2 | 6. $\frac{1}{2}$ | |